Xiran Liu

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EDUCATION AND TRAINING

Postdoc Data Science Institute & Center for Computational Molecular Biology, Brown University Oct 2023-
Advisor: Dr. Sohini Ramachandran

Ph.D. Computational and Mathematical Engineering, Stanford University Sep 2023
Dissertation: "Computational Methods and Mathematical Measures for Population Relationships"
Advisor: Dr. Noah Rosenberg

- Gene Golub Dissertation Award, *Institute for Computational and Mathematical Engineering* (2023)
- CEHG Predoctoral Fellowship, *Center for Computational, Evolutionary and Human Genomics* (2022)

M.S. Computational and Mathematical Engineering, Stanford University Sep 2020

B.S. Systems Science and Engineering, Washington University, St. Louis, *Summa Cum Laude* May 2018

B.S. Computer Science, Washington University, St. Louis, *Summa Cum Laude* May 2018

- David H. Levy Outstanding Senior Award, *Dept. of Electrical and Systems Eng.* (2018)
- Russel R. Pfeiffer Outstanding Junior Award, *Dept. of Electrical and Systems Eng.* (2017)
- Antoinette Frances Dames Award for Productive Scholarship, *School of Eng. and Applied Sci.* (2016)
- Outstanding Sophomore Award, *Dept. of Electrical and Systems Eng.* (2016)
- Dean's List, *School of Eng. and Applied Science* (2014-17)

RESEARCH INTERESTS

Computational biology and data science, with a focus on method development in population genetics, evolutionary studies, and biomedical data analysis, including devising new frameworks and data visualization tools for analyses of structured biological variation, and addressing computational challenges like characterizing the complex trait architectures using machine learning.

RESEARCH EXPERIENCE

Clustering alignment for single-cell clustering analysis, Postdoctoral Research Associate 2024-
Dr. Sohini Ramachandran, Department of Ecology, Evolution and Organismal Biology, Stanford University

- A study of clustering alignment for single cell clustering analysis to facilitate model comparison and characterization of clustering-informative genes.
- Develop a framework inspired by population-genetics clustering to align hard and mixed-membership clustering results of single cell and spatial transcriptomics data.
- Enable clustering model comparison to improve reliability and evaluate consistency of clustering results.
- Examine and interpret feature-level signals from aligned clustering results to characterize clustering-informative genes contributing to cellular heterogeneity.
- Demonstrate the framework on benchmark single-cell data and breast cancer spatial transcriptomics data.

Machine learning for multivariate genetic association analyses, Postdoctoral Research Associate 2023-25
Dr. Sohini Ramachandran, Department of Ecology, Evolution and Organismal Biology, Stanford University

- A study of advanced machine learning approaches to tackle challenges in genetic association analyses.
- Develop a computationally efficient framework, *ML-MAGES*, which leverages neural networks and variational inference for effect size shrinkage and multi-trait effect clustering in analyzing genome-wide association summary statistics while accounting for linkage disequilibrium.
- Perform gene-level analyses and summarize patterns on gene-level shared versus trait-specific associations

through visualization of multivariate high-dimension results.
- Demonstrate the applicability of method in UK Biobank data and open source the tool in Python.

Mathematical properties of genetic dissimilarity measures, Graduate Student Researcher 2022-23
Dr. Noah Rosenberg, Department of Biology, Stanford University

- A study of the distribution of the widely used allele-sharing dissimilarity measures on how their mathematical properties affect the orders and magnitudes of the empirical observations.
- Mathematically derived the inequality relationships between the expected within- and between-population measures and validated the relationships on empirical data.
- Formulated mathematical bounds on the measures with respect to the largest allele frequency and demonstrated the effects of the bounds on the distribution of empirical observations.

Cluster alignment problem in population structure analysis, Graduate Student Researcher 2021-23
Dr. Noah Rosenberg, Department of Biology, Stanford University

- A study of the alignment problem of multiple clustering results in population structure analysis.
- Devised a theoretical measure for individual contribution to misalignment cost between clustering outputs and demonstrated its usage on population ancestries inferred using HGDP microsatellites data.
- Designed a framework based on network and optimization theory, *Clumppling*, to efficiently align multiple clustering replicates with new functionalities and improved quality and efficiency compared to existing methods and implemented the program as a distributable Python package.

Dynamics of SRB and surnaming in a cultural transmission model, Graduate Student Researcher 2021
Dr. Marcus Feldman, Department of Biology, Stanford University

- A short-term course project on modeling cultural evolution.
- Simulated and analyzed a transmission model of surnaming decision and SRB (sex ratio at birth).
- Investigated the effects of several influential factors on this model, including parents' decisions, the cultural value of a daughter, reward for matrilineal surnaming, and awareness of imbalance in SRB.

Ancestry-specific association study on COVID-19 infection, Graduate Student Researcher 2020-21
Dr. Alexander Ioannidis, Biomedical Data Science & ICME, Stanford University

- Part of an institution-wide collaboration on the multi-omic pandemic tracking of COVID-19.
- Analyzed genome sequence, risk factor, and demographic data collected from Stanford Hospital.
- Examined the relationships between the severity of SARS-CoV-2 infection and local genetic ancestry and identified chromosomal segments showing potential ancestry-specific associations.

Network-based hierarchical clustering analysis on cultural variation, Graduate Student Researcher 2020
Dr. Noah Rosenberg, Department of Biology, Stanford University

- An exploratory tool to analyze culture variation using a population-genetic-originated method.
- Adapted network-based hierarchical clustering to extract informative cultural features.
- Analyzed pronunciation variation, folklore diversities, phonemic variation, and temporal naming trends.

Relationship between F_{ST} and migration in a multi-population network, Rotation Student 2019-2020
Dr. Noah Rosenberg, Department of Biology, Stanford University

- A study of the system of inferring migration patterns from pairwise F_{ST} statistics via coalescence
- Estimated migration patterns from pairwise F_{ST} using numerical solvers and examined their relationships.
- Conjectured conditions for the existence and identifiability of the system.

ADDITIONAL PROJECT EXPERIENCE

Improving fundraiser quality via natural language processing techniques, Data Scientist Intern 2022
PayPal Global Data Science Group, San Jose, CA

- Built tree-based models on text-based features and used explainable AI models to interpret the machine

learning results, providing writing suggestions for better-quality fundraiser posts.
- Trained transformer-based large language models for text classification and text generation.
- Designed and implemented guided text generation to suggest quality-improving headlines for the posts.
- Provided a baseline model with quality improvement achieved on one-third of the fundraisers and an advanced model with improvement on more than half of the fundraisers.

Visualization of phylogenetic trees in latent space using neural networks, Student Researcher 2021-22
Dr. Julia Palacios, Department of Statistics, Stanford University

- Built a supervised autoencoder framework for matrix encodings of ranked genealogies.
- Visualized ranked genealogy distribution in low-dimensional latent space.
- Differentiated α/β-splitting, demographic histories, and inferred epidemiological parameters.

Prediction of inpatient glucose levels and insulin dosing, Rotation Student 2019
Dr. Johnathan Chen, Department of Medicine - Biomedical Informatics Research, Stanford University

- Analyzed performances of human-based insulin management in electronic health records.
- Built a prediction framework of inpatient glucose levels and insulin dosing using machine learning.

Tumor classification using imaging data of diffusion MRI, Undergraduate Researcher 2017-18
Dr. Jr-Shin Li, Department of Electrical and Systems Engineering, Washington University in St. Louis

- Using image processing techniques to process and align the histology and diffusion MRI imaging data of brain tumor and prostate cancer to automate data labeling.
- Built a classification framework to differentiate tumor pathologies using supervised machine learning.

Optimization of quality-cost trade of human annotations for webpage data, Research Intern 2016
IPAM-RIPS 2016 Program, Institute for Pure and Applied Mathematics, UCLA

- Simulated human-annotated webpage data and compared techniques to mitigate human errors in classifying webpage contents using supervised machine learning.
- Proposed optimal strategies to use budge efficiently in carrying out human-annotation tasks.

PUBLICATIONS

Peer-Reviewed Journal Articles

Liu X, Crawford L, Ramachandran S (2025). ML-MAGES enables multivariate genetic association analyses with genes and effect size shrinkage. *Genome Res. (accepted)*

Liu X, Ahsan Z, Rosenberg NA (2025). Using mathematical constraints to explain narrow ranges for allele-sharing dissimilarities. *Theor. Popul. Biol*. *(in press)*

Liu X, Kopelman N, Rosenberg NA (2024). Clumppling: cluster matching and permutation program with integer linear programming. *Bioinformatics* 40(1).

Liu X, Ahsan Z, Martheswaran T, Rosenberg NA (2023). When is the allele-sharing dissimilarity between two populations exceeded by the allele-sharing dissimilarity of a population with itself?. *Stat. Appl. Genet. Mol. Biol.*

22(1).

Liu X, Kopelman N, Rosenberg NA (2022). A Dirichlet model of alignment cost in mixed-membership unsupervised clustering. *J. Comput. Graph. Stat.* 1-35.

Liu X, Rosenberg NA, Greenbaum G (2022). Extracting hierarchical features of cultural variation using network-based clustering. *Evol. Hum. Sci.* 4.

Parikh VN, Ioannidis AG, Jimenez-Morales D, Gorzynski JE, De Jong HN, Liu X, Roque J, Cepeda-Espinoza VP, *et al.* (2022) Deconvoluting complex correlates of COVID-19 severity with local ancestry inference and viral phylodynamics: Results of a multi-omic pandemic tracking strategy. *Nat. Comm. 13*(1), 1-10.

Liu X, Feldman MW (2021). Effects of cultural transmission of surnaming decisions on the sex ratio at birth. *Theor. Popul. Biol*. *141*, 44-53.

Ye Z, Price RL, Liu X, *et al.* (2020). Diffusion histology imaging detects and classifies glioblastoma pathology missed by conventional magnetic resonance imaging. *Clin. Cancer Res. 26*(20):5388-5399.

Work in Progress

Liu X, Singh R, Ramachandran S. Clustering alignment for single-cell clustering analysis facilitates model comparison and characterization of clustering-informative genes contributing to cellular heterogeneity. *(in prep)*

Ahsan Z, Liu X, Rosenberg NA. Combinatorics of a dissimilarity measure for pairs of draws from discrete probability vectors on finite sets of objects. (*under review; preprint on arXiv*)

Liu X, Greenbaum G, Rosenberg NA. Can migration be inferred from pairwise F_{ST}? *(in prep)*

Application Notes

Guo A, Ramachandran S, Liu X. *KAlignedoscope*: an interactive visualization tool for aligned clustering results. *(under review)*

Liu X, Rosenberg NA, Ramachandran S. *Clumppling 2.0*: A Clustering Alignment Program for Population Structure Analyses. *(in prep)*

CONFERENCE PRESENTATIONS

"Clustering alignment for single cell analyses: streamlining model comparison and revealing informative genes." **Oral presentation** at *Genome Informatics 2025*. Cold Spring Harbor Laboratory, NY, November 2025.

"Clustering alignment for single cell analysis facilitates model comparison and characterization of informative genes contributing to cellular heterogeneity." **Poster presentation** at *The American Society of Human Genetics 2025 Annual Meeting*. Boston, MA, October 2025.

"*ML-MAGES:* A machine learning framework for multivariate genetic association analyses with genes and effect size shrinkage." **Oral presentation** at *The Research in Computational Molecular Biology 2025 Conference*. Seoul, Korea, April 2025.

"*ML-MAGES*: a computationally efficient tool for multivariate genetic association analyses with effect size shrinkage via machine learning." **Poster presentation (Reviewers' Choice Abstract - top 10%)** at *The American Society of Human Genetics 2024 Annual Meeting*. Denver, CO, November 2024.

"Mathematical properties of allele-sharing dissimilarities." **Poster presentation** at *The Allied Genetics Conference 2024*. Metro Washington, DC, March 2024.

"*Clumppling*: a new method for aligning replicate solutions in population structure analysis." **Oral presentation** at *Evolution 2023 Conference*. Albuquerque, NM, June 2023.

"A Dirichlet model of alignment cost in mixed-membership clustering results of ancestry inference." **Oral presentation** at *Bay Area Population Genomics Conference*. UC Berkeley, Berkeley, CA, October 2022.

"Modeling alignment cost in mixed-membership unsupervised genetic clustering." **Poster presentation** at *Population, Evolutionary, and Quantitative Genetics Conference*. Pacific Grove, CA, May 2022.

"Optimizing the quality-cost trade-off of human annotation for labeling webpages to train webpage classifiers." (*Sucar E*, **Liu X***, Khillan S*, Li Z**) **Poster presentation** at *Nebraska Conference for Undergraduate Women in Mathematics*. Lincoln, NE, Feb 2017.

"Optimizing the quality-cost trade-off of human annotation for labeling webpages to train webpage classifiers." (*Khillan S*, Sucar E*, **Liu X***, Li Z**) **Conference abstract accepted** for AMS Contributed Paper Session on Undergraduate Research at *Joint Mathematics Meetings*. Atlanta, GA, Feb 2017.

TEACHING EXPERIENCE

Undergraduate Research Mentor, Ramachandran Lab, *Brown University* 2025-

- Co-supervise an undergraduate student in summer research program on advancing interactive visualization for clustering alignment in population genetics.
- Guide the student in project design, methodological thinking, communication, and scientific writing.
- Encourage independent learning and professional growth, leading toward a manuscript in preparation.

DSCoV Workshop Instructor, Data Science Institute, *Brown University* 2025
Data Science, Computation, and Visualization

Instructor, Short Course, *Stanford University*
Advanced MATLAB for Scientific Computing (CME 292) Winter 2022, 2023

- A 4-week short course for students of all levels interested in advanced MATLAB features and toolboxes.
- Designed course syllables, prepared materials, and delivered lectures both over Zoom and in class.
- Open-sourced teaching materials on MATLAB File Exchange with >1.2K downloads (as of Oct 2024).

Graduate Teaching Assistant, *Stanford University*
Linear Algebra with Application to Engineering Computations (CME 200/ME 300A)
Dr. Parviz Moin, Department of Mechanical Engineering, Stanford University Fall 2021
Dr. Margot Gerritsen, ICME, Stanford University Fall 2020
Overall effectiveness rating in course evaluation: average of 4.31/5 from 161/173 responses over 2 offerings

- A lecture-style course for approximately 90 graduate students in engineering.
- Held office hours twice a week and led recitation sessions.
- Designed exercise problems and graded problem sets, coding projects, and exams.

Ordinary Differential Equations for Engineers (CME 102) 2020-22
Dr. Hung Le, ICME, Stanford University
Overall effectiveness rating in course evaluation: average of 4.26/5 from 97/133 responses over 3 offerings

- A lecture-style course for approximately 50 undergraduate students in engineering.
- Held office hours twice a week and led recitation sessions.
- Designed exercise problems and graded problem sets, coding projects, and exams.

Summer Workshop on Intro to Machine Learning, *ICME* 2021

Peer Mentor, *Washington University in St. Louis* 2015-17

Undergraduate Teaching Assistant, *Washington University in St. Louis* 2015-17
Engineering Freshman Seminar (EN 120, Course Assistant)
Algorithms and Data Structure (CSE 241/247)
Introduction to Electrical Engineering (ESE 103)
Signals and Systems (ESE 351)

LEADERSHIP AND SERVICE

Ad hoc Reviewer: *PNAS*, *Theor. Popul. Biol.*

Reviewer (GSA Journals Peer Review Training Program), *Genetics* 2021-23

Ambassador and Organizing Committee Member, *Women in Data Science (WiDS) at Brown University* 2024-

- Co-organized the Women in Data Science (WiDS) Providence 2024 Conference.
- Planed the event agenda and invited speakers for talks and panels.
- Brought over 100 students and community members for a day of panels, seminars, and networking events.

International Chair, *Brown Postdoc Council* 2024-

- Organized social and networking events for postdoctoral researchers.

Reviewer, 2024 *SACNAS* NDiSTEM Conference 2024

Member, *ICME Student Action Group, Stanford University* 2021-23

- Proposed improvements for the ICME program, discussed proposals with faculty and staff, and monitored the progress of proposal implementation.
- Resolved concerns in the general student body and organized student community events.

Member and Secretary, *Alpha Omega Epsilon Sorority Beta Xi Chapter, Washington University* 2016-18

Program Assistant, *International Summer Study Programs, Washington University* 2015, 2017

Volunteer, *Campus Y, Washington University* 2015-16

SKILLS

Programming Languages: Python, R, MATLAB, C++, MySQL, Mathematica, JavaScript, HTML

Software, Packages, and Tools: Git, Slurm, PyTorch, TensorFlow, Hugging Face, PLINK, BEAST 2

Languages: Chinese, English, German (basic)